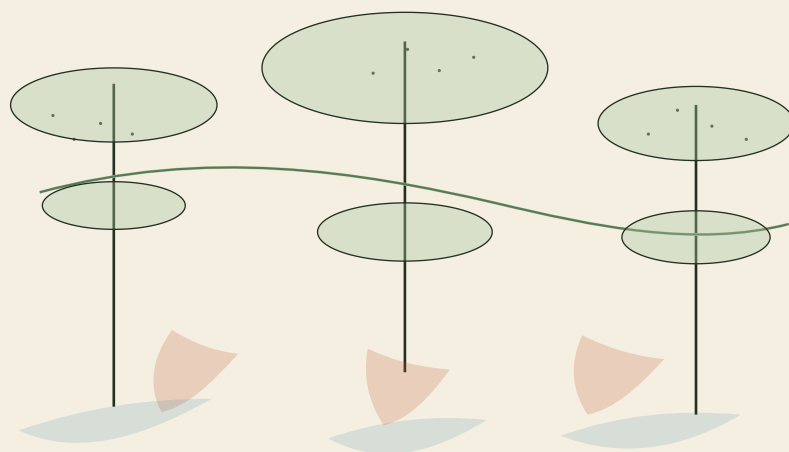




nabta  WELLNESS PARK

THE LIVING SHADE LOOP

DUBAI MUNICIPALITY • AI PARK DESIGN CHALLENGE • DRAFT COMPETITION ENTRY

D·07

Landscape, Public Realm, Sustainability & Resilience

№07 Landscape

№09 Sustainability



Deliverable 7 — Landscape & Public Realm Plans + Sustainability & Resilience Strategy

«نبّة» NABTA — Wellness Park. The Living Shade Loop. The landscape is the infrastructure: an 858 m continuous shaded network — a 398 m figure-eight braided into a 460 m perimeter ring, roughly 1 km including the wadi spur — threads ten garden rooms across the irregular 15,001 m² parcel (a 16-vertex polygon, ~120 × 178 m, street edge to the south). Canopy density in each room is proposed by AI solar and UTCI simulations and checked room-by-room by the landscape team against the six personas — resident-panel confirmation is planned before design freeze — AI drafts, humans decide. Where thermal stress peaks, the canopy thickens; where evening breezes pass, it opens. *Nabta means "sprout": the brand is the planting strategy itself — the park opens as ≈450 sprouting native saplings and visibly grows into its own identity year on year.*

(A) LANDSCAPE STRATEGY — THREE PLANTING TIERS

- **Tier 1 — Canopy (≈450 native trees):** ghaf, sidr and samr form the shade spine, with date palms as cultural accents at the gateways. Simulation-driven targets range from 35% canopy cover in the event lawn room to 85% in the quiet contemplation garden and picnic groves.
- **Tier 2 — Understorey shrubs:** drought-adapted flowering shrubs at 0.6–1.8 m height frame garden rooms, screen the loop edges and feed pollinators.
- **Tier 3 — Groundcover:** desert grasses and salt-tolerant mats replace lawn everywhere except the 2,000 m² multipurpose event lawn, cutting turf area by roughly 80% against a conventional district park.

Sensory & therapeutic layer. As a wellness park, two quiet rooms are planted as therapy gardens: fragrant vitex and Texas sage at seated nose height, rustling thumam grass for calming white noise, smooth gabbro touch-walls, and a barefoot texture path (sand, granite fines, timber). Raised, wheelchair-height planting beds host horticultural-therapy sessions with local clinics; bloom is staggered so scent and colour shift monthly, giving repeat visitors a reason to return.

BOTANICAL NAME	COMMON / ARABIC	WATER NEED	PRIMARY ROLE
Prosopis cineraria	Ghaf / غاف	Very low	Shade, biodiversity
Ziziphus spina-christi	Sidr / سدر	Low	Shade, pollinators
Vachellia tortilis	Samr / سمّر	Very low	Shade, nesting habitat
Phoenix dactylifera	Date palm / نخيل	Moderate	Shade, cultural identity
Leucophyllum frutescens	Texas sage	Low	Sensory (bloom, scent)
Vitex agnus-castus	Chaste tree / شجرة إبراهيم	Low	Sensory (scent), pollinators
Panicum turgidum	Thumam / ثمام	Very low	Sensory (sound), seed for birds
Pennisetum divisum	Desert fountain grass	Very low	Bioswale stabilisation
Sesuvium portulacastrum	Sea purslane	Low (salt-tolerant)	Groundcover, lawn substitute
Tetraena qatarensis	Harm / هرم	Very low	Groundcover, biodiversity

All canopy and groundcover species are UAE-native; Texas sage and chaste tree are the only introduced species — long proven in low-water Gulf plantings and confined to the two sensory rooms.

(B) PUBLIC REALM MATERIALS

Paths Light-coloured, high solar-reflectance stabilised decomposed granite for the loop and secondary trails — permeable, cool underfoot, wheelchair-firm, and locally sourced.

Shade structures Corten steel frames with FSC timber slats form pergolas over rest nodes, weathering without paint or maintenance coatings.

Seat walls Local UAE stone (gabbro and sandstone) seat walls double as room edges and thermal mass shaded by canopy. All fixed lighting is 2700K LED on solar-hybrid poles with adaptive dimming, targeting 60% lighting energy reduction versus code baseline.

(C) WATER STRATEGY & BUDGET

A wadi-inspired bioswale runs beneath the loop's low edge, harvesting AC condensate piped from adjacent villas plus rare storm flows into a 150 m³ underground cistern. TSE tops up demand. Sub-surface drip with soil-moisture sensors and an AI evapotranspiration controller irrigates by tier, not by timer. Target: **41% below conventional park irrigation** (105 → 62 m³/day at peak; supply balances exactly: 9 condensate + 53 TSE = 62).

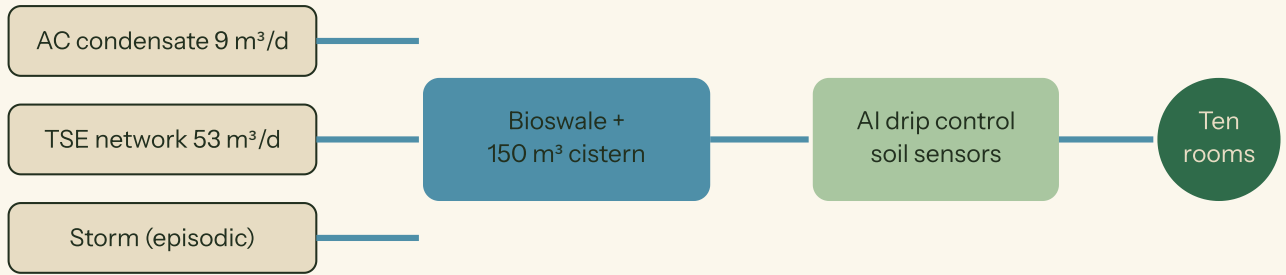
WATER BUDGET (PEAK SUMMER)	M³ / DAY	NOTES
Conventional benchmark demand	105	Turf-dominant park of equal area (15,001 m²)
NABTA design demand	62	Native tiers + drip + sensors (-43 m³ = -41%)
Supply: AC condensate (villas)	9	≈120 units × 75 L/day, summer
Supply: TSE network	53	Balance of demand
Supply: stormwater to cistern	episodic	Offsets shoulder-season TSE draw

(D) BIODIVERSITY & CLIMATE RESILIENCE

Targets by year 3: 25+ resident and migratory bird species (purple sunbird, laughing dove, hoopoe, white-eared bulbul), 15+ pollinator species supported by staggered ghaf, sidr and vitex bloom, and log-and-stone habitat piles in two quiet rooms. All Tier 1–3 species are verified against a 2050 scenario (+2°C, higher salinity, longer dry spells): every canopy species already thrives in interior UAE conditions hotter than Dubai's projected climate, and the salt-tolerant groundcovers hedge against declining TSE quality. No permanent potable irrigation connection is installed.

(E) SOCIAL & ECONOMIC SUSTAINABILITY

A 120 m² café concession at the southern street-edge gateway and bookable event-lawn slots (markets, fitness, school programmes) generate an estimated AED 250–350k/year toward operations (~70% of upkeep). Low-water tiers, no-mow groundcovers and self-finishing corten target a maintenance cost of **≤ AED 28/m²/year (≈ AED 420k/year for 15,001 m²)** against a conventional turf-park benchmark of AED 50–60/m²/year: the 80% turf reduction removes most mowing and fertiliser labour, sensor-led drip cuts irrigation callouts, and the figure includes a 5% annual plant-replacement allowance plus a corrective-works sinking fund. Years 1–3 carry a higher establishment budget (staking, formative pruning, sapling watering) before settling to steady state; a community planting programme with local schools builds stewardship and trims replanting cost. Within the AED 35M budget, landscape and irrigation works are capped at 38%, protecting contingency for shade structures.



WATER FLOW: HARVESTED AND RECYCLED SOURCES FEED SENSOR-LED DRIP IRRIGATION – 62 M³/DAY, 41% BELOW THE CONVENTIONAL BENCHMARK.